Friday: Independent Project - Week 1

Agenda

1. **Instructions**
   - Learning Outcomes
   - Deliverables
2. **Assessment**
   - Part 1: Career Planning
   - Part 2: Python Programming
   - Part 3: SQL Programming
3. **Evaluation**

Instructions

During this week's Independent project, you will get to test the skills that you learned throughout this week's sessions. More specifically, you will get the test your understanding of the following learning outcomes.

Overall Learning Outcomes

- I can understand what it takes to become a Data Scientist.
- I can recall the basics of Python programming for data science.
- I can obtain and manipulate data from various types of databases using the SQL language.

Deliverables

The deliverables for this week's Independent project include:

1. A write up outlining your Data Science career plan.
2. A Python Colaboratory Notebook with your responses.
3. An SQL Colaboratory Notebook with your responses.

Assessment

Part 1: Career Planning

Overview

In this part of the assessment, you will be required to write a career development plan for your data science career. This will involve recording, reflecting, tracking, planning and reviewing the steps that you will take for the next five years of your data science career. Your career plan can take into account of any career path that you would want to undertake i.e. employment, entrepreneurship, consulting, etc.

Deliverable

1. You will be required to provide a write up (Google Docs Document) outlining your Data Science career plan. You can use the learnings from this week's sessions.
2. Use the following naming convention for the above Google Docs Document.
   - "Moringa_Data_Science_Prep_W1_Independent_Project_2019_06_FirstName_LastName_Career_Plan"
3. Have the above document put in one folder that you will submit together with the other deliverables of this Independent Project.

Part 2: Python Programming

Overview

During this week, you also started learning how as a Data Scientist, you will be required to write programs that perform python computations.

In this section, you will be required to create several python programs that should be able to perform the given operations then, display the results of the operations.

- Take in two values from a user (both in pounds) then convert those values to kilograms.
- Perform the sum of the values.
- Perform the average of the values.
- Find the difference between both values.
- Find the quotient when one value is divided by the other.

- Determine and print out whether any of the numbers are even or add.
- Have comments applied appropriately.

Deliverable

1. You will create a Colaboratory Notebook that will contain your responses to the above questions.
2. Use the following naming convention for the above Google Colaboratory Notebook,
   - "Moringa_Data_Science_Prep_W1_Independent_Project_2019_06_FirstName_LastName_Python_Notebook"
3. Do not clear your Notebook consoles upon completion of this deliverable.
4. Have the above document put in one folder that you will submit together with the other deliverables of this Independent Project.

Part 3: SQL Programming

Overview

In this section, you will act as a Data Science Consultant who will answer questions posed using a dataset collected by Dalberg. The dataset contains crops grown in Uganda.

SQL Programming Questions

1. Display a list of Sub Counties and their population and areas.
2. Sort the list of districts by total crop area (descending order).
3. Select only the Sub counties from the Moroto district, order them alphabetically and show their production of sorghum.
4. Compute the total Maize production per District.
5. Compute the number of Sub counties where Maize is produced and the total Maize production per District.
6. Compute the overall Crop area in all Sub counties where population is over 20000.
7. Sort the Maize production in descending order by Districts, only taking into account Sub counties where Maize area is larger than Sorghum area, and display the number of Sub counties per district matching that criteria.

Dataset Description

This dataset contains yield and population per subcounty. The dataset for the above questions can be found here. [Link **(https://drive.google.com/a/moringaschool.com/file/d/1pWXDvs33OoULTH4kdzhGTUxJmp0dSgZq/view?usp=sharing)**].

The glossary for this table is as follows:

- POP: total population for the subcounty
- S_Yield_Ha: average yield for sorghum for the subcounty (Kg/Ha)
- M_Yield_Ha: average yield for maize for the subcounty (Kg/Ha)
- Crop_Area_Ha: total crop area for the subcounty (Ha)
- S_Area_Ha: total sorghum crop area for the subcounty (Ha)
- M_Area_Ha: total maize crop area for the subcounty (Ha)
- S_Prod_Tot: total productivity for the sorghum for the subcounty (Kg)
- M_Prod_Tot: total productivity for the maize for the subcounty (Kg)

Deliverable

1. You will create an Colaboratory Notebook that will contain your responses to the above questions.
2. Use the following naming convention for the above Notebook:
   - "Moringa_Data_Science_Prep_W1_Independent_Project_2019_06_FirstName_LastName_SQL_Notebook"
3. All questions, text responses, SQL queries should be included in the Notebook.
4. Do not clear your Notebook consoles upon completion of this deliverable.
5. Put the file(s) in one folder then follow the given instructions to make the submission.

Submission & Evaluation

The submission to this week's Independent Project should be made here [**Link (https://moringaschool.instructure.com/courses/156/assignments/1658)**]. This submission will be a link to the Google Drive folder that contains three files. You should also ensure that you set the sharing permissions of the folder to "Anyone with the link can view".

A rubric will be used for evaluation the above assessment.

Please note:

1. The deadline for this assessment is 6.00pm on the day of the assessment.
2. **Important**: Please make sure to set the sharing permissions of the folder to "Anyone with the link can view" or else your work won't be accessible by your Technical Mentor.
3. Late submissions or without full editing access and assessments that have been edited after the deadline will not be assessed.

4. Do not seek to copy someone else's work while working on this Independent project. You deny yourself an opportunity to learn whenever you resolve to plagiarism.

"The goal is to turn data into information, and information into insight." - Carly Fiorina